

Solving Quadratic Equations by Factoring

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A **quadratic equation** is an equation that can be written in the form

$$ax^2 + bx + c = 0,$$

where $a \neq 0$.

Solving by Factoring

To solve a quadratic equation by **factoring**:

Write the equation in **standard form**.

Factor the quadratic expression completely.

Apply the **Zero Product Property**.

Solve each resulting linear equation.

Zero Product Property

If $AB = 0$, then $A = 0$ or $B = 0$.

Greatest Common Factor

Some quadratic equations require finding the **Greatest Common Factor (GCF)** before factoring.

$$2x^2 + 10x = 0$$

Factor out the GCF:

$$2x(x + 5) = 0$$

Checking Solutions

Always **check** each solution by substituting it into the original equation.

Original equation



Substitute each solution



Verify

Examples

Example 1

Solve by factoring.

$$x^2 + 7x + 12 = 0$$

Factor:

$$(x + 3)(x + 4) = 0$$

By the Zero Product Property:

$$x + 3 = 0 \quad \text{or} \quad x + 4 = 0$$

$$x = -3 \quad \text{or} \quad x = -4$$

Example 2

Solve by factoring.

$$2x^2 + 10x = 0$$

Factor out the GCF:

$$2x(x + 5) = 0$$

By the Zero Product Property:

$$2x = 0 \quad \text{or} \quad x + 5 = 0$$

$$x = 0 \quad \text{or} \quad x = -5$$

Example 3

Solve by factoring.

$$2x^2 + 7x + 3 = 0$$

Factor:

$$(2x + 1)(x + 3) = 0$$

By the Zero Product Property:

$$2x + 1 = 0 \quad \text{or} \quad x + 3 = 0$$

$$x = -\frac{1}{2} \quad \text{or} \quad x = -3$$

Practice Exercises

Practice 1

Solve the quadratic equation by factoring.

$$x^2 + 7x + 12 = 0$$

Practice 2

Solve the quadratic equation by factoring.

$$x^2 - 9x + 20 = 0$$

Practice 3

Solve the quadratic equation by factoring.

$$x^2 - 16 = 0$$

Practice 4

Solve the quadratic equation by factoring.

$$2x^2 + 10x = 0$$

Practice 5

Solve the quadratic equation by factoring.

$$3x^2 - 12x = 0$$

Practice 6

Solve the quadratic equation by factoring.

$$2x^2 + 7x + 3 = 0$$

Practice 7

Solve the quadratic equation by factoring.

$$6x^2 - x - 2 = 0$$

Practice 8

Solve the quadratic equation by factoring.

$$4x^2 - 25 = 0$$

Practice 9

Solve the quadratic equation by factoring.

$$5x^2 + 15x + 10 = 0$$

Activities

Activity 1

Solve the quadratic equation by factoring.

$$x^2 + 9x + 20 = 0$$

Activity 2

Solve the quadratic equation by factoring.

$$x^2 - 11x + 24 = 0$$

Activity 3

Solve the quadratic equation by factoring.

$$x^2 - 49 = 0$$

Activity 4

Solve the quadratic equation by factoring.

$$2x^2 + 14x = 0$$

Activity 5

Solve the quadratic equation by factoring.

$$4x^2 - 20x = 0$$

Activity 6

Solve the quadratic equation by factoring.

$$3x^2 + 8x + 4 = 0$$

Activity 7

Solve the quadratic equation by factoring.

$$2x^2 - 7x + 3 = 0$$

Activity 8

Solve the quadratic equation by factoring.

$$9x^2 - 64 = 0$$

Activity 9

Solve the quadratic equation by factoring.

$$4x^2 + 20x + 16 = 0$$

Thank you!